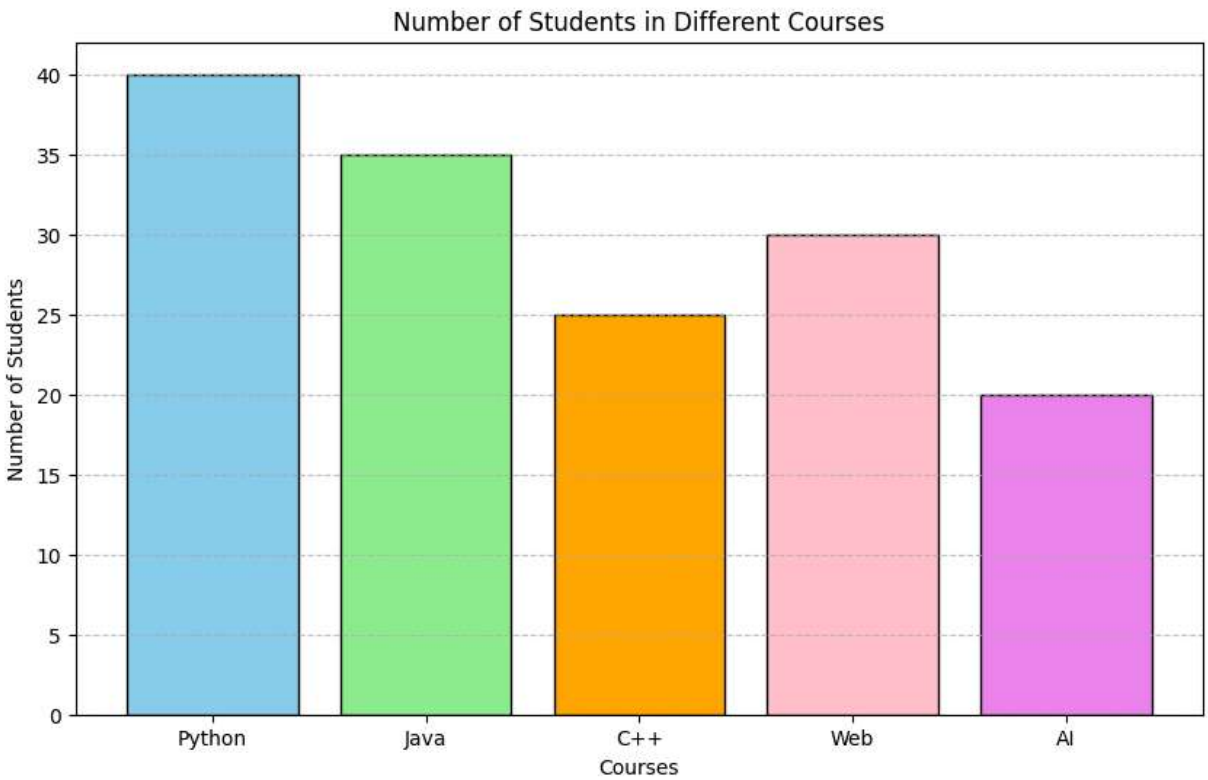
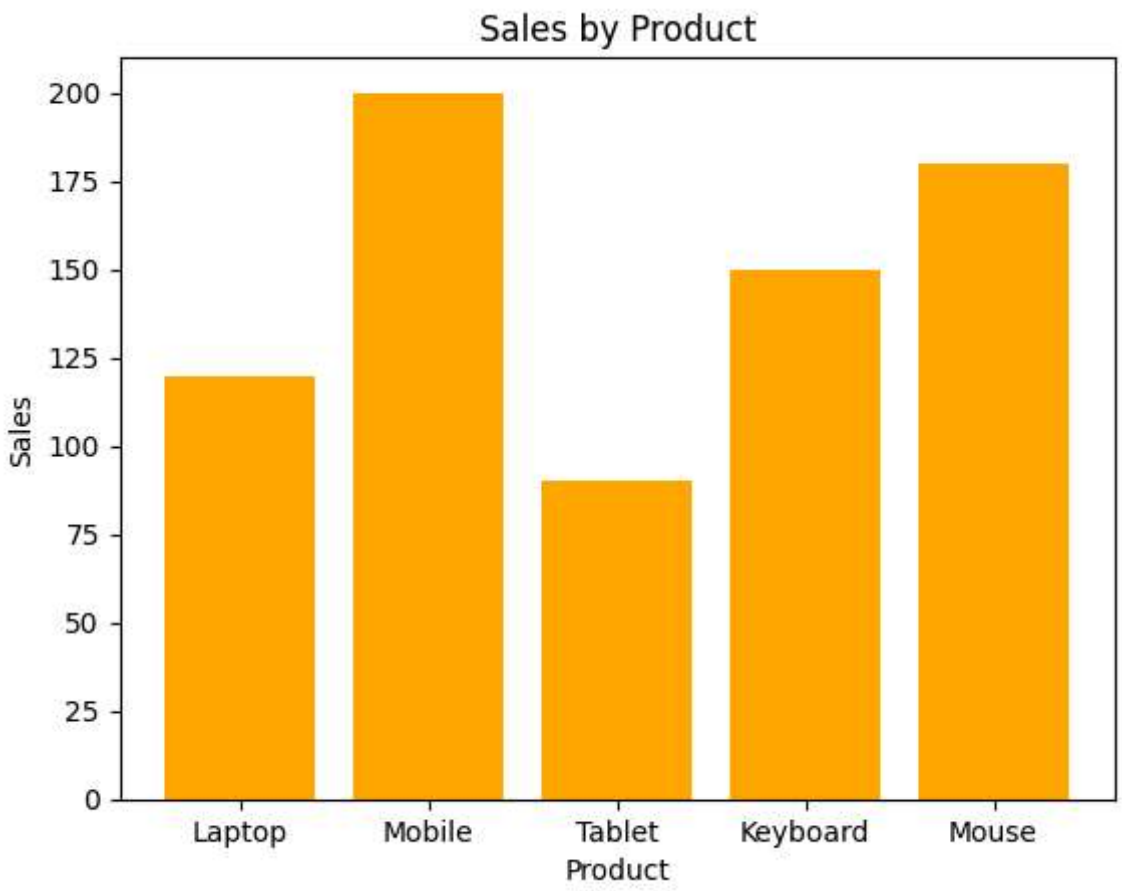
In [1]:
```
import matplotlib.pyplot as plt

# Data
courses = ["Python", "Java", "C++", "Web", "AI"]
students = [40, 35, 25, 30, 20]

# Create Bar Chart
plt.figure(figsize=(10,6))
plt.bar(courses, students, color=['skyblue', 'lightgreen', 'orange', 'pink', 'viole
plt.title("Number of Students in Different Courses")
plt.xlabel("Courses")
plt.ylabel("Number of Students")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

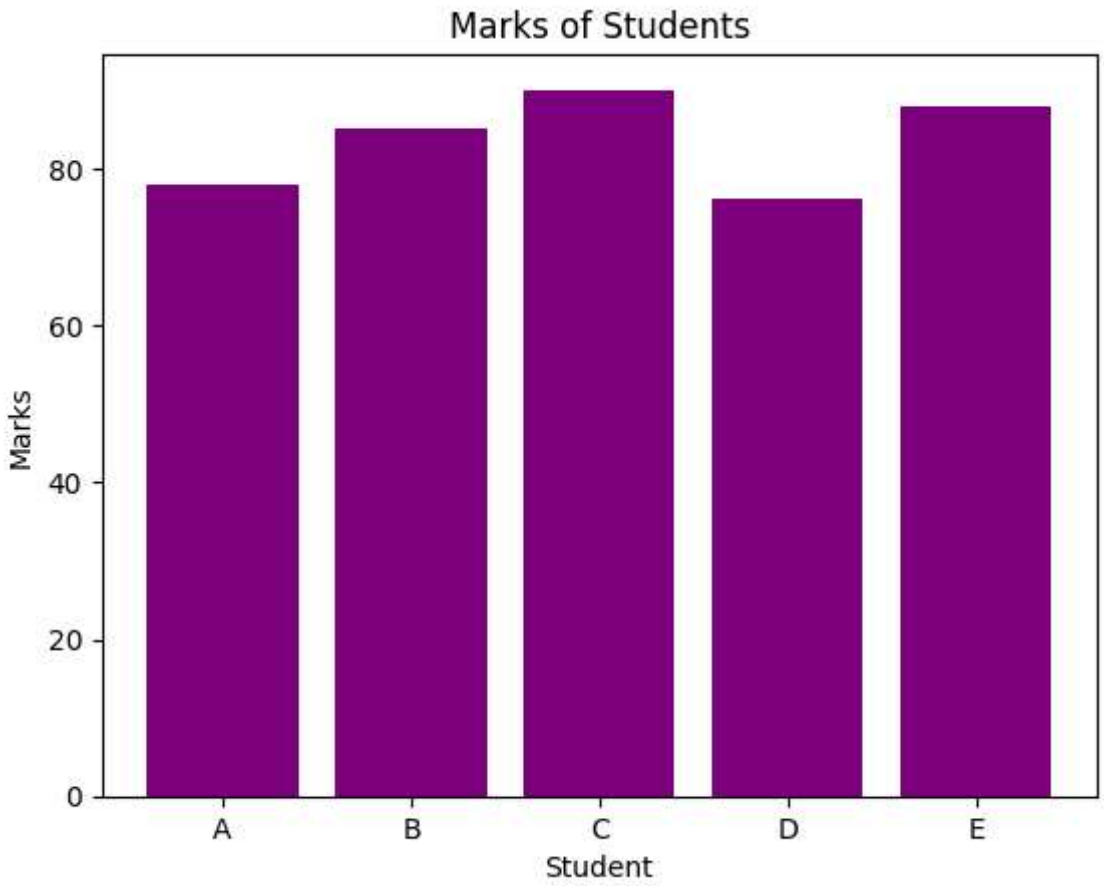


In [2]:
```
import matplotlib.pyplot as plt

# Data
products = ['Laptop', 'Mobile', 'Tablet', 'Keyboard', 'Mouse']
sales = [120, 200, 90, 150, 180]

# Create bar chart
plt.bar(products, sales, color='orange')

# Add title and labels
plt.title('Sales by Product')
plt.xlabel('Product')
plt.ylabel('Sales')
```

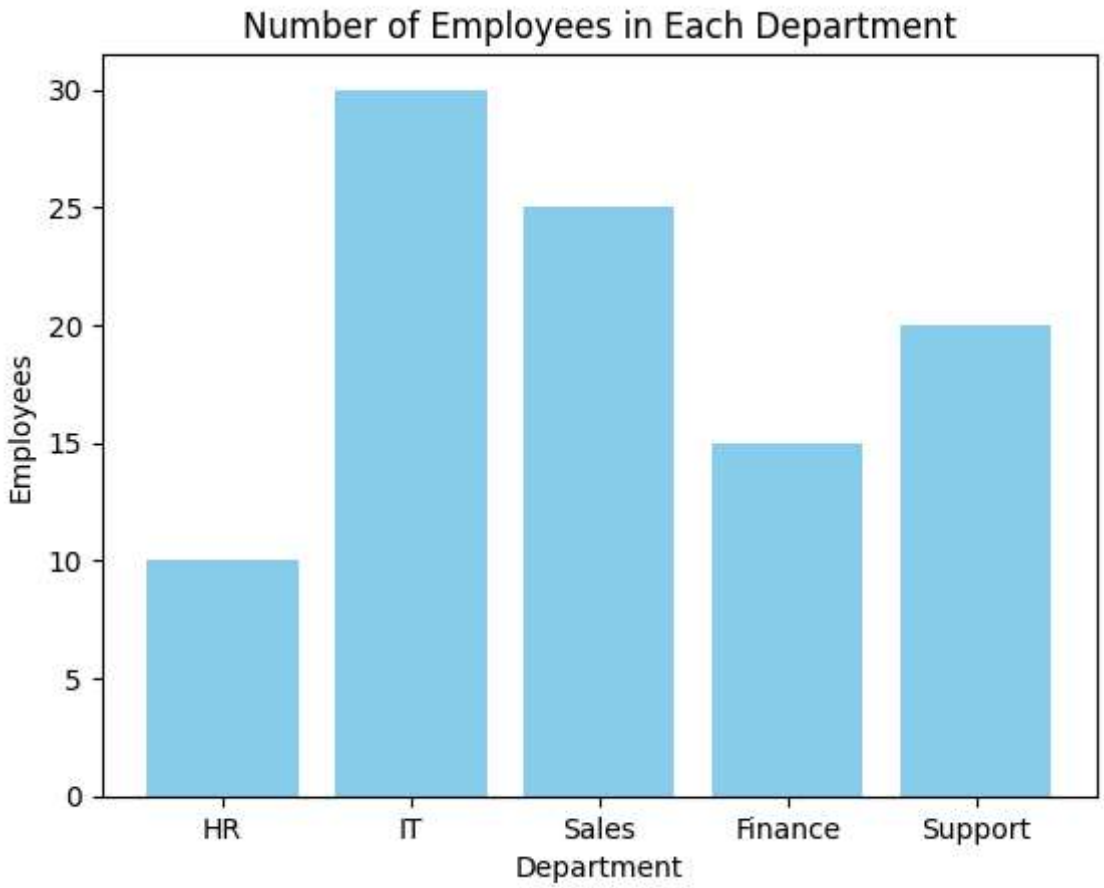
```
# Show chart
plt.show()
```



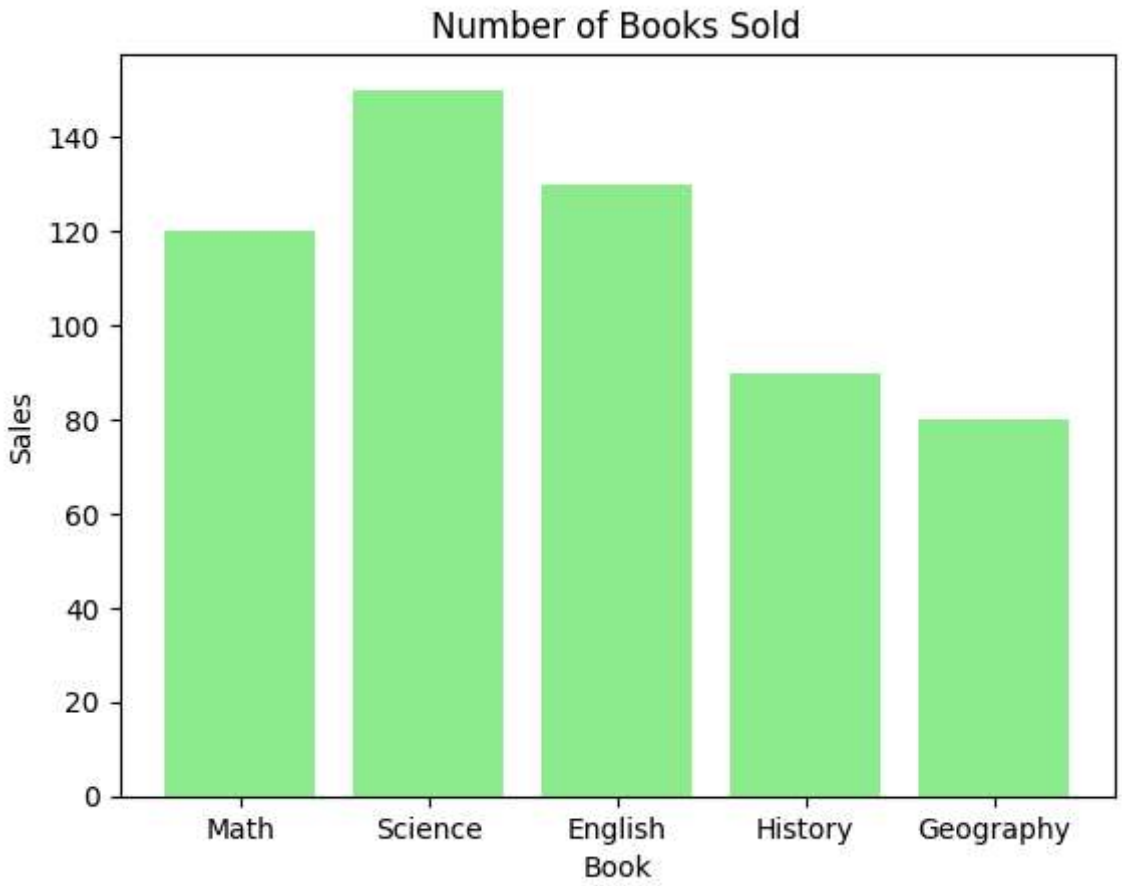
In [3]:
```
import matplotlib.pyplot as plt

# Data
students = ['A', 'B', 'C', 'D', 'E']
marks = [78, 85, 90, 76, 88]

# Create bar chart
plt.bar(students, marks, color='purple')

# Add title and labels
plt.title('Marks of Students')
plt.xlabel('Student')
plt.ylabel('Marks')

# Show chart
plt.show()
```



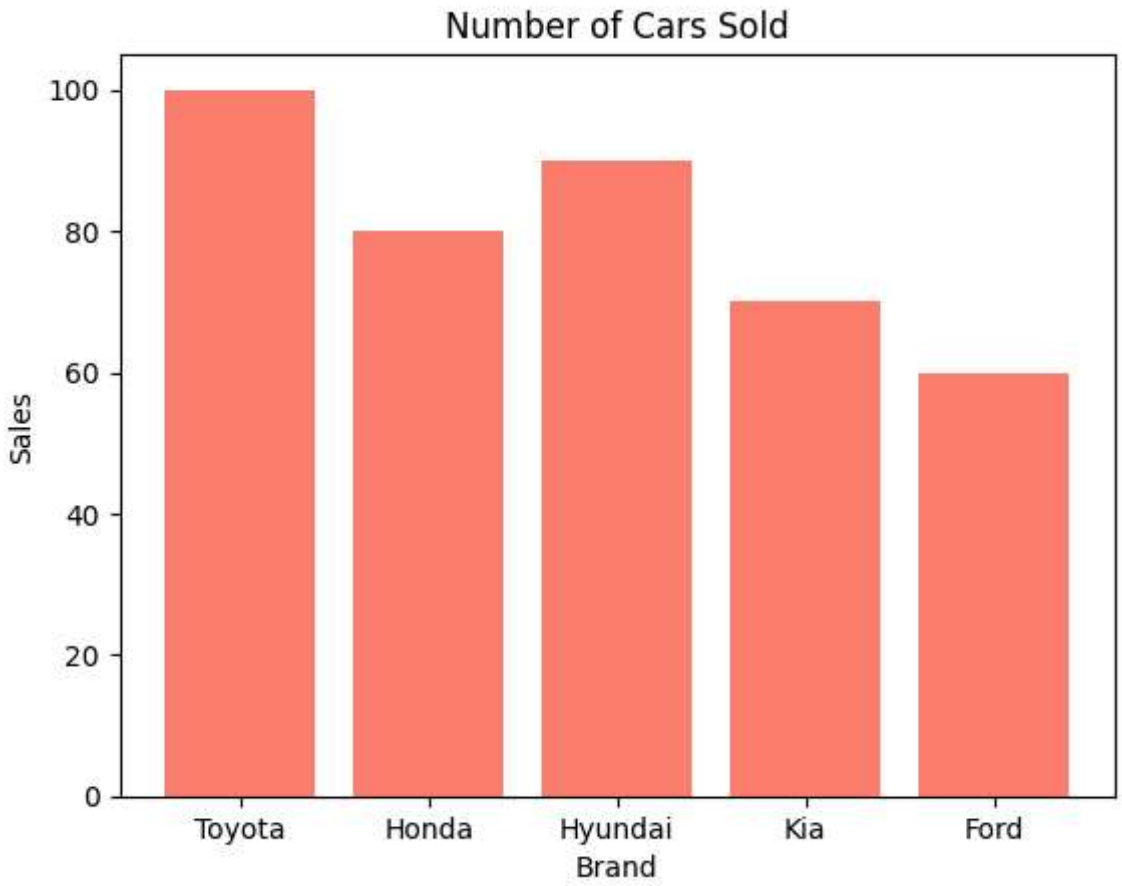
In [4]:
```
import matplotlib.pyplot as plt

# Data
departments = ['HR', 'IT', 'Sales', 'Finance', 'Support']
employees = [10, 30, 25, 15, 20]

# Create bar chart
plt.bar(departments, employees, color='skyblue')
plt.title('Number of Employees in Each Department')
plt.xlabel('Department')
plt.ylabel('Employees')
plt.show()
```



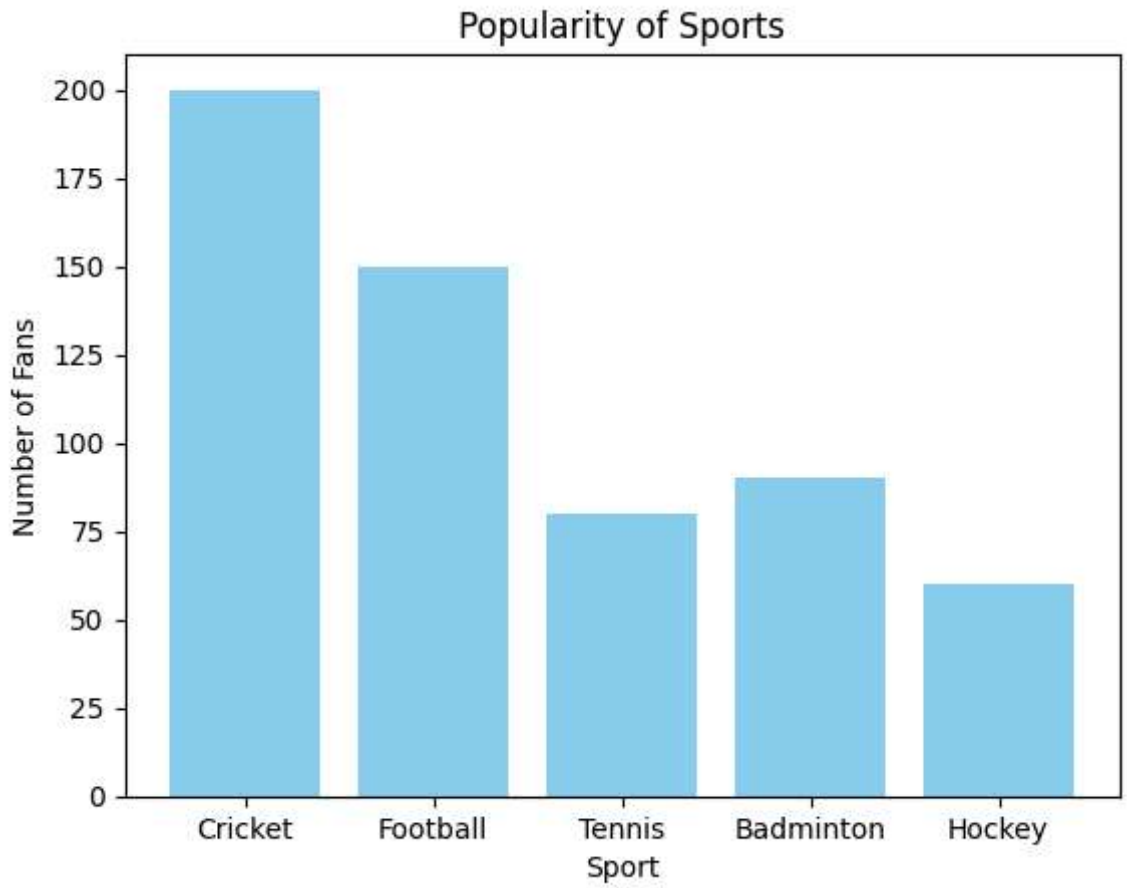
In [5]:
```
import matplotlib.pyplot as plt

# Data
books = ['Math', 'Science', 'English', 'History', 'Geography']
sales = [120, 150, 130, 90, 80]

# Create bar chart
plt.bar(books, sales, color='lightgreen')
plt.title('Number of Books Sold')
plt.xlabel('Book')
plt.ylabel('Sales')
plt.show()
```



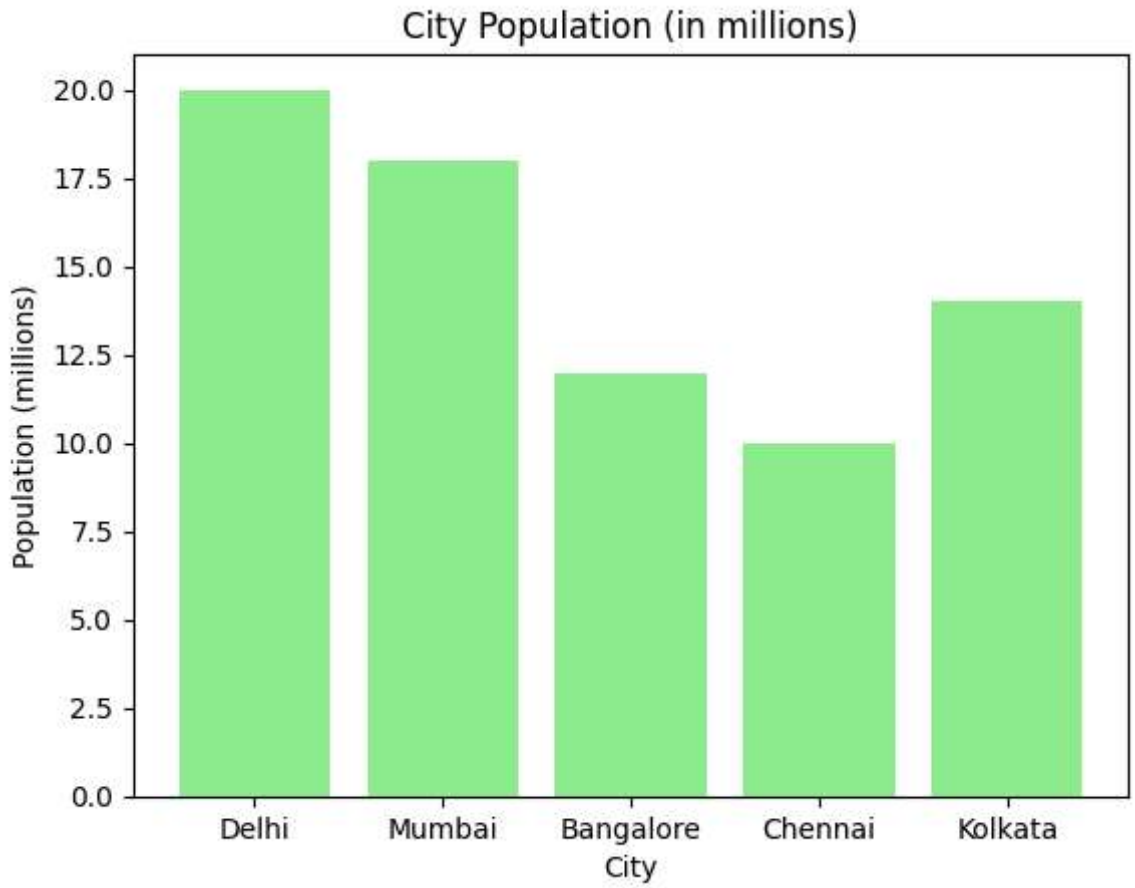
In [7]:
```
import matplotlib.pyplot as plt

# Data
brands = ['Toyota', 'Honda', 'Hyundai', 'Kia', 'Ford']
sales = [100, 80, 90, 70, 60]

# Create bar chart
plt.bar(brands, sales, color='salmon')
plt.title('Number of Cars Sold')
plt.xlabel('Brand')
plt.ylabel('Sales')
plt.show()
```



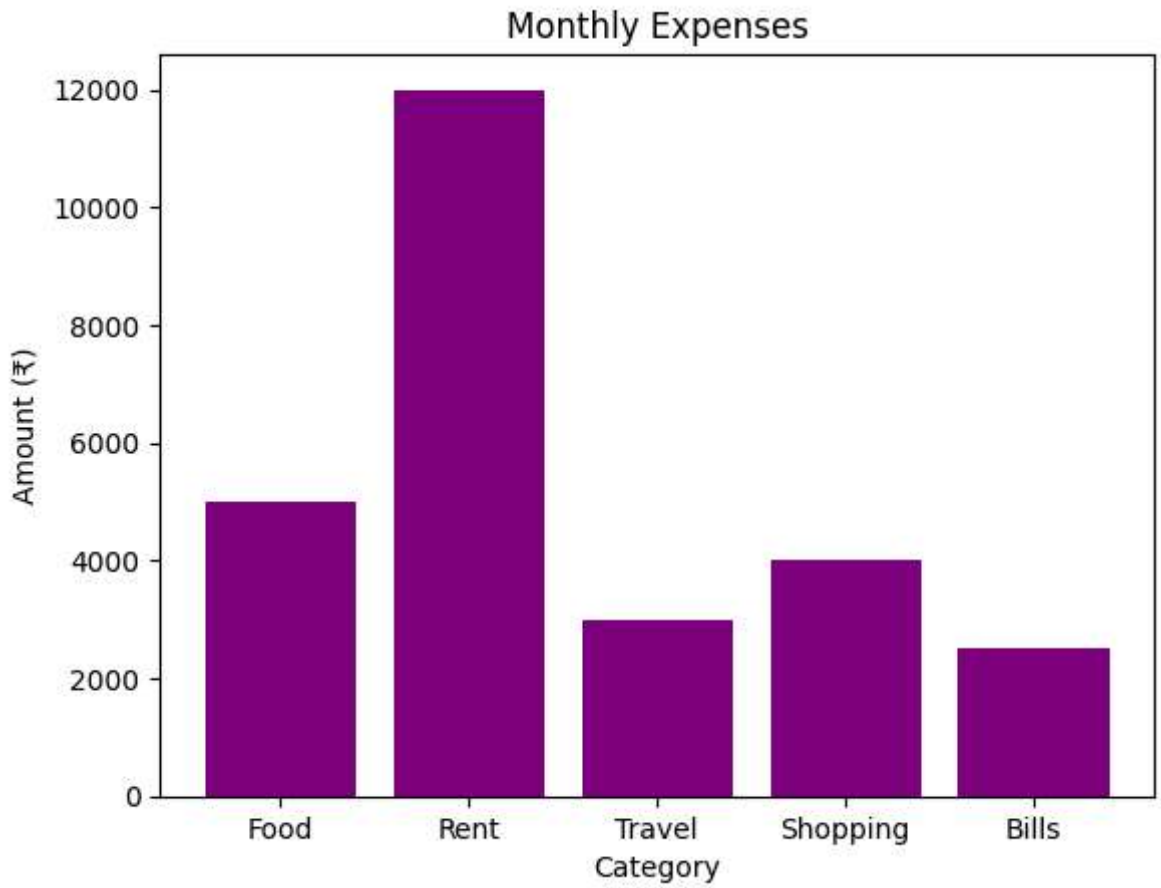
In [8]:
```
import matplotlib.pyplot as plt

# Data
sports = ['Cricket', 'Football', 'Tennis', 'Badminton', 'Hockey']
fans = [200, 150, 80, 90, 60]

# Create bar chart
plt.bar(sports, fans, color='skyblue')
plt.title('Popularity of Sports')
plt.xlabel('Sport')
plt.ylabel('Number of Fans')
plt.show()
```



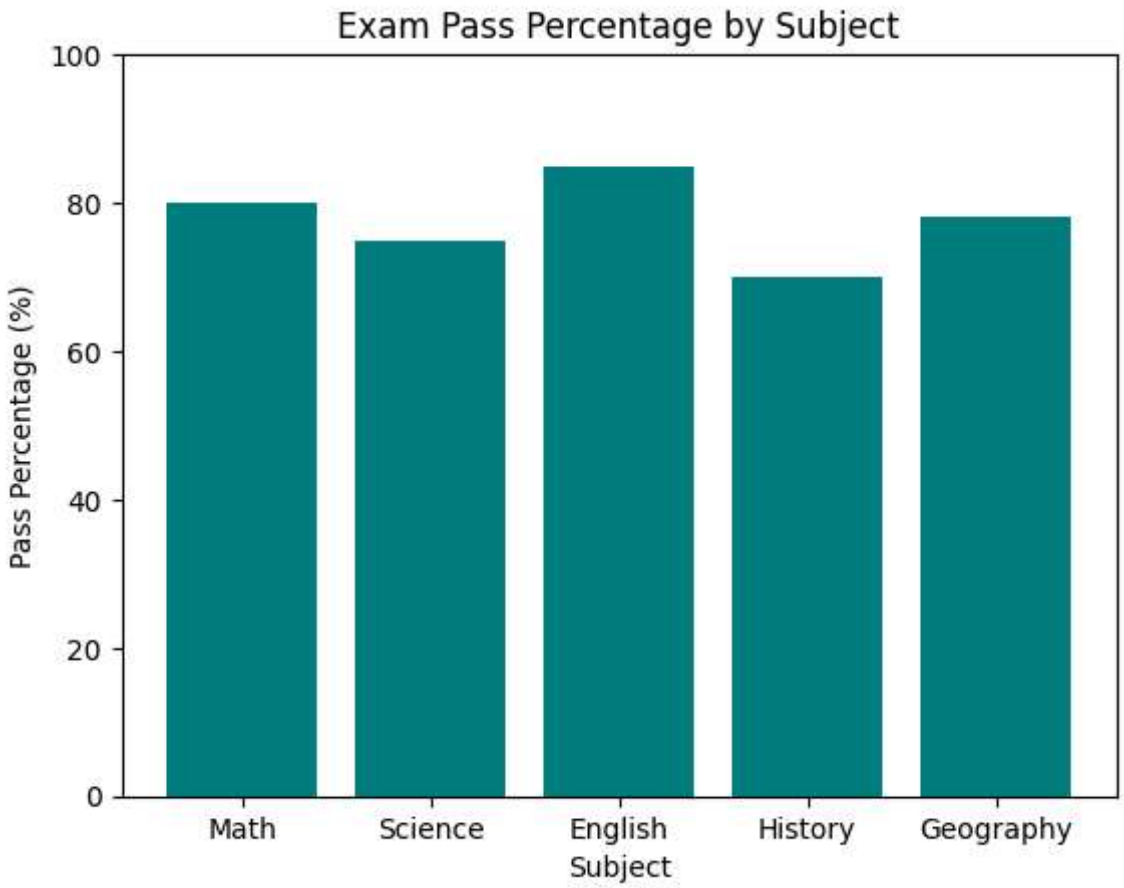
In [9]:
```
import matplotlib.pyplot as plt

# Data
cities = ['Delhi', 'Mumbai', 'Bangalore', 'Chennai', 'Kolkata']
population = [20, 18, 12, 10, 14]  # in millions

# Create bar chart
plt.bar(cities, population, color='lightgreen')
plt.title('City Population (in millions)')
plt.xlabel('City')
plt.ylabel('Population (millions)')
plt.show()
```



In [10]:
```
import matplotlib.pyplot as plt

# Data
categories = ['Food', 'Rent', 'Travel', 'Shopping', 'Bills']
amounts = [5000, 12000, 3000, 4000, 2500]

# Create bar chart
plt.bar(categories, amounts, color='purple')
plt.title('Monthly Expenses')
plt.xlabel('Category')
plt.ylabel('Amount (₹)')
plt.show()
```



In [11]:
```
import matplotlib.pyplot as plt

# Data
subjects = ['Math', 'Science', 'English', 'History', 'Geography']
pass_percentage = [80, 75, 85, 70, 78]

# Create bar chart
plt.bar(subjects, pass_percentage, color='teal')
plt.title('Exam Pass Percentage by Subject')
plt.xlabel('Subject')
plt.ylabel('Pass Percentage (%)')
plt.ylim(0, 100)  # Since percentage ranges from 0 to 100
plt.show()
```


Exam Pass Percentage by Subject
Pass Percentage (%)
100
80
60
40
20
0
Math
Science
English
History
Geography
Subject

In [ ]: